JAWWAD AHMAD

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EDUCATION

Jamia Millia Islamia — *New Delhi*
B.Tech in Computer Science Engineering (Data Science) — Honours in Applied Intelligence — *2024 — Present*
Overall CGPA: 8.67 | 4th Semester SGPA: 8.80

Senior Secondary (XII), CBSE — Science — *St. Michael's High School, Patna*
2023 — *Percentage: 95.00%*

EXPERIENCE

FOSSEE Semester Internship (Extension) — OSDAG Software ↗ — *Feb 2026 — Apr 2026*
Indian Institute of Technology (IIT), Bombay
- Awarded a merit-based extension to lead the public-release integration of **OsBridgeLCCA**.
- Improved **computational efficiency**, data validation, and reliability of structural-design software modules.

FOSSEE Winter Internship — OSDAG Software Domain ↗ — *Dec 2025 — Jan 2026*
Indian Institute of Technology (IIT), Bombay
- Developed **OsBridgeLCCA** for life-cycle cost modeling and analysis of numerical structural-design data.
- Automated **2D CAD drawing generation** using ezDXF and built visualization workflows with **Matplotlib and xarray**.

SOFTWARE ENGINEERING PROJECTS

myMachino — Construction Equipment Marketplace ↗
Tech Stack: Next.js, TypeScript, PostgreSQL, Drizzle ORM, Zod, Tailwind CSS
- Built and deployed a **mobile-first equipment marketplace** connecting customers with construction and commercial equipment service providers.
- Developed customer requirement workflows capturing **equipment, location, quantity, date, duration, operator requirements, and budget**.
- Designed an admin-driven **lead management and provider database** with status tracking, equipment availability, search/filtering, and provider assignment.
- Implemented the core **customer → lead → provider → booking** workflow with structured validation and responsive UI.

LedgerPilotAI — AI Finance Controller ↗
Tech Stack: React, TypeScript, Node.js, Express, PostgreSQL, NeonDB, Drizzle ORM
- Built a production-oriented **financial reconciliation platform** processing payment and ledger records using exact matching, fuzzy scoring, and AI-assisted matching.
- Implemented **audit trails, human-in-the-loop review**, and natural-language queries for explainable cash-position analysis.

USBot — Offline Edge AI & Efficient Inference ↗
Tech Stack: Python, PySide6, Ollama, FastAPI, Vector Stores
- Engineered a **privacy-centric offline AI assistant** for local inference targeting latency, bandwidth, and privacy constraints.
- Optimized local inference pipelines for **model-state management and low-latency interactions** on consumer hardware.
- Implemented local **RAG** with embeddings and semantic search for on-device document intelligence.

PUBLICATIONS

An Efficient OS-Inspired Retrieval-Centric Approach for Offline LLM Inference on Resource-Constrained Edge Devices ↗ — ***Acceptance Letter*** ↗
Accepted *at the Springer International Conference on Next Generation Communication and Computing (**NGCCOM 2026**); to be published in Springer LNNS (**Scopus Indexed**).*

TECHNICAL SKILLS

Languages:	Python, C++, Java, JavaScript, TypeScript, SQL
Software Development:	Next.js, React, Node.js, Express, FastAPI, REST APIs, PostgreSQL, Drizzle ORM, Zod, Tailwind CSS
AI & Machine Learning:	RAG, Vector Databases, Local LLM Inference, LangChain, LangGraph, Ollama, PyTorch, Pandas, NumPy, Scikit-learn
Information Retrieval:	Dense Retrieval, BM25, Reciprocal Rank Fusion (RRF), Reranking, ColBERTv2, FAISS, Embeddings, Context Expansion
Systems & Tools:	KV-Cache Optimization, SQL Optimization, Schema Design, Git/GitHub, Docker, Linux, Figma, Streamlit, Matplotlib, NeonDB

ACHIEVEMENTS

Research Extension:	Awarded a semester extension at IIT Bombay based on technical performance.
Odoo Hackathon Finalist (2026):	Ranked **479 out of 25,000+ applicants**.
Competitive Programming:	Solved **100+ problems** on LeetCode.